

L8: Case Study: Metabolic Networks

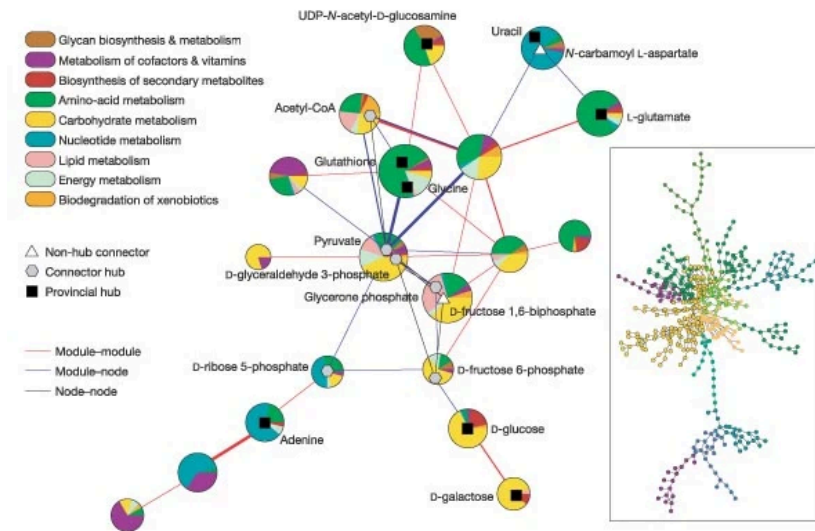


Figure 3: Cartographic representation of the metabolic network of *E. coli*. from [Functional cartography of complex metabolic networks](https://www.nature.com/articles/nature03288) by Guimerà, R., Nunes Amaral, L.

To illustrate these concepts, let us review the main results of a research article (*“Functional cartography of complex metabolic networks”* by Guimerà and Luis A. Nunes Amaral, *Nature* 2005) that analyzed the metabolic networks of twelve organisms. They identified an average of 15 communities (*the maximum was 19 communities for Homo sapiens and the minimum was 11 for Archaeoglobus fulgidus*).

The visualization shows the network of these communities in the case of *E. coli* (19 communities). Each circle corresponds to a community. The colors within each community are associated with a specific type of metabolic pathways, classified based on the KEGG database (see upper left).

The triangles, hexagons, or squares within some communities identify specific metabolites that play the role of non-hub connectors, connector hubs, or provincial hubs, respectively. For example, L-glutamate is a provincial hub within the amino-acid metabolism module.

The network at the right shows the *E. coli* metabolic network (473 nodes, 574 edges, using the KEGG color code).